

[◆ DATA ARCHITECTURE](#)

KINETIC TYPE

Backend Schema Document

Local-first IndexedDB schema, its Firestore mirror, security rules, and the sync contract that keeps both in agreement.

[INDEXEDDB / DEXIE](#)[CLOUD FIRESTORE](#)[SECURITY RULES](#)

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01 Storage Architecture Overview

IndexedDB is the permanent source of truth on every device. Cloud Firestore is a mirror — used purely for backup and cross-device continuity — and is never in the critical path of a live typing session.



02 IndexedDB — Local Schema

TABLE: NGRAM_STATS

FIELD	TYPE	DESCRIPTION
pair_id	String (PK)	Compound key string, e.g. <code>e->r</code> , <code>Shift->{</code>
domain	Enum	<code>CODE_TS</code> , <code>CODE_PY</code> , <code>CLI_BASH</code> , <code>PROSE</code> — pairs are tracked per-domain
avg_latency_ms	Float	EWMA transition time, $\lambda = 0.15$
total_occurrences	Integer	Total times typed
error_count	Integer	Uncorrected miss-hits
weight_wp	Float	Cached last-computed W(P) selection weight
last_updated	Timestamp	Unix epoch of last record

TABLE: SESSION_LOGS

FIELD	TYPE	DESCRIPTION
session_id	UUID (PK)	Client-generated unique session identifier
domain_type	Enum	<code>CODE_TS</code> , <code>CODE_PY</code> , <code>CLI_BASH</code> , <code>PROSE</code> , <code>MEDICAL</code>
wpm_raw	Float	Gross speed ((Total Characters / 5) / Minutes)
wpm_net	Float	Net speed accounting for uncorrected errors
burst_consistency	Float	Variance score of speed consistency over 5-second intervals
top_pairs_json	String (JSON)	Snapshot of top-5 targeted pairs and their session-local latency
timestamp	Timestamp	Session completion time

TABLE: SRS_QUEUE NEW

FIELD	TYPE	DESCRIPTION
pair_id	String (PK, FK → ngram_stats)	Pair under spaced-repetition tracking
mastered_at	Timestamp	When the pair first dropped below the "weak" threshold
next_review_at	Timestamp	Next scheduled low-density re-insertion
review_stage	Integer	Spaced-repetition interval stage (0–5)

03 Cloud Firestore — Schema

COLLECTION: USERS/{UID}

FIELD	TYPE	DESCRIPTION
display_name	String null	Set only after account upgrade
created_at	Timestamp	First anonymous sign-in time
is_anonymous	Boolean	False once linked to a real credential
settings	Map	{ theme, adaptivity_level, reduced_motion, default_domain }
last_synced_at	Timestamp	Most recent successful sync flush

SUBCOLLECTION: USERS/{UID}/NGRAM_STATS/{PAIR_ID}

Mirrors the local table exactly, keyed identically by pair_id so merges are a direct field-level operation — no ID translation layer.

SUBCOLLECTION: USERS/{UID}/SESSIONS/{SESSION_ID}

Append-only mirror of session_logs, keyed by the same client-generated UUID for idempotent upserts.

TOP-LEVEL COLLECTION: CORPORA/{DOMAIN_ID}

FIELD	TYPE	DESCRIPTION
domain_id	String (PK)	CODE_TS, CLI_BASH, PROSE, ...
version	Integer	Corpus manifest version, bumped on content update
source_manifest_url	String	Hosting CDN path to the versioned corpus JSON bundle
license_notes	String	Attribution for public-domain / open-source sourced text

Public, read-only, served independent of any user — see Security Rules §06.

04 Entity Relationships



CARDINALITY NOTES

- One user has exactly one `ngram_stats` row per (`pair_id`, `domain`) combination — a pair's stats are tracked independently per domain, since "Shift→{" behaves differently in Dev Mode vs. Prose Mode context.
- `session_logs` are immutable and never updated after creation — corrections are handled by writing a superseding row, never an in-place edit, preserving a clean audit trail for trend charts.
- `srs_queue` entries are derived, not independently authored — they are created/removed by the worker's spaced-repetition logic reacting to `ngram_stats` changes.

05 Indexes

STORE	INDEX	PURPOSE
IndexedDB <code>ngram_stats</code>	<code>[domain+weight_wp]</code>	Fast retrieval of top-N weighted pairs per domain for the synthesizer
IndexedDB <code>session_logs</code>	<code>[domain_type+timestamp]</code>	History/trend queries scoped by domain and date range
Firestore <code>ngram_stats</code>	<code>domain ASC, weight_wp DESC</code>	Composite index for potential future leaderboard/percentile features
Firestore <code>sessions</code>	<code>timestamp DESC</code>	History pagination, most recent first

06 Security Rules

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    // Users can only ever read/write their own subtree
    match /users/{uid}/{document=**} {
      allow read, write: if request.auth != null
        && request.auth.uid == uid;
    }

    // Corpora are public, read-only, writable only by Cloud Functions
    match /corpora/{domainId} {
      allow read: if true;
      allow write: if false; // admin SDK only
    }
  }
}
```

07 Sync & Conflict Resolution

TABLE	CONFLICT STRATEGY
ngram_stats	Weighted-average merge on <code>avg_latency_ms</code> , weighted by each side's <code>total_occurrences</code> ; <code>total_occurrences</code> and <code>error_count</code> sum directly.
session_logs	Append-only + idempotent upsert by <code>session_id</code> — no merge logic needed, duplicates are naturally deduplicated.
srs_queue	Not synced directly — recomputed locally per device from the (already-merged) <code>ngram_stats</code> .
users/{uid}.settings	Last-write-wins by <code>last_synced_at</code> — acceptable since settings are low-frequency, single-intent writes.

WHY NOT LAST-WRITE-WINS EVERYWHERE

Naively overwriting `ngram_stats` would let a stale device silently erase real progress recorded on another device. The occurrence-weighted merge guarantees both devices' history is preserved in the combined average.

08 Data Retention & Privacy

- Raw keystroke *content* (which characters were typed, beyond what's needed to render the session in progress) is never persisted or synced — only derived timing/error metrics.
- Local data persists indefinitely until the user clears browser storage or explicitly deletes it via Settings.
- Cloud data is retained for the lifetime of the account; account deletion triggers a Cloud Function that purges the full `users/{uid}` subtree.
- Anonymous accounts with no activity for 180 days are eligible for automated cleanup (nightly Cloud Function job).

09 Sample Documents

```
// users/8f2c.../ngram_stats/Shift->{
{
  "pair_id": "Shift->{",
  "domain": "CODE_TS",
  "avg_latency_ms": 146.8,
  "total_occurrences": 312,
  "error_count": 9,
  "weight_wp": 220.7,
  "last_updated": 1798761600
}
```

```
// users/8f2c.../sessions/3e91...
{
  "session_id": "3e91-...-uuid",
  "domain_type": "CODE_TS",
  "wpm_raw": 108.2, "wpm_net": 104.0,
  "burst_consistency": 0.91,
  "top_pairs_json": "[{\"pair\":\"Shift->{\", \"ms\":142}]",
  "timestamp": 1798765200
}
```